

SUPERVISOR'S REVIEW

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Class: Advanced Program: Power System and Renewable Energy 01 - K67

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1. Thesis topic

Resilience-Oriented Peer-to-Peer Energy Trading in Multi-Microgrids via Network-Constrained Distributed Predictive Control.

2. Review

Regarding presentation format:

- The thesis has a logical layout; the chapters and sections are tightly organized. The visual presentation of the text is beautifully formatted, highly aesthetic, and includes all the necessary elements required for a standard graduation thesis.
- The figures are sharp, and the charts are clear with detailed annotations, well illustrating the research results.
- The figures are sharp, and the charts are clear with detailed annotations, well illustrating the research results.
- The writing style strictly adheres to the academic standards of a graduation thesis. The accurate and consistent use of specialized terminology throughout the manuscript reflects the student's serious dedication to the research.

Regarding content:

- The research is highly innovative by successfully combining three layers: spatial, physical, and temporal. It utilizes SOCP relaxation for the physical grid model, spatial decoupling via the ATC algorithm to secure information among agents, and model predictive control (MPC) for real-time optimization.
- The simulation calculations and results are directly relevant to and closely aligned with the thesis objectives. Through extreme fault scenarios, the computational data demonstrates that the system has an excellent capability to self-heal and maintain stability. The student has proven the reliability of the distributed algorithm when facing severe operating conditions, thereby demonstrating outstanding independent research capabilities and problem-solving skills.
- The thesis holds profound scientific significance and is highly practical for power grid disaster response, aligning closely with current advanced research

trends. The achieved results provide a solid mathematical foundation for resilient microgrid designs in the future.

- Although the model has shown outstanding results in various fault scenarios, there are still challenges to be addressed to enhance its practical applicability. Notably, the current algorithm does not account for the impacts of communication delays, which are highly prevalent in large-scale distributed control systems.

Therefore, future research should focus on developing an Asynchronous ATC mechanism to overcome communication delays, while also deploying the model on a Hardware-in-the-Loop (HIL) real-time system to comprehensively evaluate the feasibility of the solution.

Hanoi, 24 June, 2026

SUPERVISOR

Assoc. Prof. Nguyen Duc Tuyen